AP ELECTRIC · AP FOCO: PCA9685 0x60 (CTRL), INA238 0x4C/0x4D (corriente + ID), 2 x AL8860.
Rset del foco: I [mA] = 5000 / R [kOhm] (5 k = 1 A, 10 k = 500 mA, 14.3 k = 350 mA).

VIN_RAW PWR_FLAG
GND PWR_FLAG
VCC PWR_FLAG
3V3 PWR_FLAG

J1 QWIIC ENTRADA
J2 QWIIC SALIDA
U1 PCA9685PW
C1 100nF
C2 10uF
JP1 +1
RA1 10k
J3 12–24V
F1 3A
D1 SS34
D2 SMBJ26A
C3 10uF 50V
TP1 VCC
TP2 GND
U4 AL8860MP
RSC1 0R1
L1 47uH
D3 SS34
C11 4.7uF 50V
RC1 2.2k
U2 INA238
C13 100nF
JP2 DIR
RID1 10k 1%
CID1 100nF
J4 FOCO 1
U5 AL8860MP
RSC2 0R1
L2 47uH
D4 SS34
C12 4.7uF 50V
RC2 2.2k
U3 INA238
C14 100nF
JP3 DIR
RID2 10k 1%
CID2 100nF
J5 FOCO 2

La corriente la fija la resistencia del foco o la app
Placa de 2 capas, 1 oz (JLCPCB)
PCB 60 x 56 mm, 2 capas

Sheet: /
File: AP_FOCO.kicad_sch

Title: AP ELECTRIC · AP FOCO – 2 focos de corriente constante adaptable

Size: A3	Date: 2026-10-11	**Rev: 1.0**
KiCad E.D.A. 9.0.9		Id: 1/1